Note

adj A is a square matrix of order n and $\text{adj } A = \left[A_{ij}\right]^T = \left[(-1)^{i+j} M_{ij}\right]^T$.

In particular, adj A of a square matrix of order 3 is given below:

$$\text{adj } A = \begin{bmatrix} +M_{11} & -M_{12} & +M_{13} \\ -M_{21} & +M_{22} & -M_{23} \\ +M_{31} & -M_{32} & +M_{33} \end{bmatrix}^T = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^T = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}.$$

Theorem 1.1

For every square matrix A of order n, $A(\text{adj } A) = (\text{adj } A)A = |A| I_n$.

Proof

For simplicity, we prove the theorem for $n = 3$ only.

Consider $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$. Then, we get

$$a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13} = |A|, \quad a_{11}A_{21} + a_{12}A_{22} + a_{13}A_{23} = 0, \quad a_{11}A_{31} + a_{12}A_{32} + a_{13}A_{33} = 0;$$
$$a_{21}A_{11} + a_{22}A_{12} + a_{23}A_{13} = 0, \quad a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23} = |A|, \quad a_{21}A_{31} + a_{22}A_{32} + a_{23}A_{33} = 0;$$
$$a_{31}A_{11} + a_{32}A_{12} + a_{33}A_{13} = 0, \quad a_{31}A_{21} + a_{32}A_{22} + a_{33}A_{23} = 0, \quad a_{31}A_{31} + a_{32}A_{32} + a_{33}A_{33} = |A|.$$

By using the above equations, we get

$$A(\text{adj}A) = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix} = \begin{bmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{bmatrix} = |A| \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = |A| I_3 \qquad \ldots (1)$$

$$(\text{adj}A)A = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{bmatrix} = |A| \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = |A| I_3, \qquad \ldots (2)$$

where I_3 is the identity matrix of order 3.

So, by equations (1) and (2), we get $A(\text{adj } A) = (\text{adj } A)A = |A| I_3$. ■

Note

If A is a singular matrix of order n, then $|A| = 0$ and so $A(\text{adj } A) = (\text{adj } A)A = O_n$, where O_n denotes zero matrix of order n.

Example 1.1

If $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$, verify that $A(\text{adj } A) = (\text{adj } A)A = |A| I_3$.

Solution

We find that $|A| = \begin{vmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{vmatrix} = 8(21-16) + 6(-18+8) + 2(24-14) = 40 - 60 + 20 = 0.$

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